Literature Survey & Research Gap Analysis: Hallucination Detection in LLMs

***Abstract*—This document presents a comparative analysis focused on detecting hallucinations in Large Language Models (LLMs). It analyses methodologies such as semantic entropy, alongside comprehensive surveys and factuality evaluation taxonomies. Furthermore it identifies computational research gaps and maps the proposed resolution to the United Nations Sustainable Development Goal 9 (Industry, Innovation and Infrastructure).**

***Index Terms*—Deep Learning, Hallucination Detection, Large Language Models, Semantic Entropy, SDG 9.**

I. Methodology

The foundational paper for this survey is "Detecting hallucinations in language models using semantic entropy" [1]. The authors propose a method to address one of the most challenging tasks in natural language processing: distinguishing a model's linguistic uncertainty from factual uncertainty.

The earlier approaches make use of Entropy, a metric that tries to capture a model's confusion by looking at responses, and seeing if changing the prompt causes the model to generate different tokens. Semantic Entropy fixes this flexibility issue; A language model can produce multiple answers, all expressing the same factual meaning (e.g., "Paris" and "It is Paris"). Naively, these are considered to have different factual meaning, but actually express the same fact.

The idea of semantic entropy is to cluster outputs according to their meaning, and then perform calculations similar to Entropy. By using entailment checks, the algorithm produces "meaning clusters", and if an AI is able to consistently produce answers with the same meaning, its semantic entropy would be low and confident. On the other hand, if a model has a high semantic entropy, that means it is very unsure about its answer, and is likely hallucinating.

II. Research Gap Identification

Upon reviewing the literature, a research gap pertaining to efficiency was noted. In particular, hallucination detection methods are often too slow to be used in production.

To calculate semantic entropy [1], one has to sample responses (typically 10+ answers per prompt) and perform additional Natural Language Inference classification or independent verifications to calculate Entropy. This multipass approach is prohibitively expensive for most applications, especially for time-critical systems like customer service chatbots. Additionally, these methods provide no guarantees in cases where a model has confidently learned false information and consistently outputs the same incorrect answers. In such cases, calculating Entropy would erroneously imply that a model is confident and factual.

The proposed mini-project focuses on the following research question: there is a need for a lightweight hallucination detection architecture that can be used in real-time applications without sampling multiple model responses, while providing strong theoretical guarantees in detecting confidently wrong responses.

III. Mapping to Sustainable Development Goals

The research conducted as part of this assignment, along with the proposed future mini-project, contribute in significant ways to the objectives of the United Nations.

Modern industries are starting to adopt Large Language Models as part of their digital infrastructures, for data processing, decision making, and customer interactions. However, the possibility for hallucination presents an obstacle to reliable and sustainable industrial innovation. A faulty AI system can cause downstream applications to behave incorrectly, causing failures in infrastructures [2], [3].

By addressing the hallucination detection research gap, this project helps provide the reliability necessary to secure sustainable industrial innovation. Securing the integrity of AI models enables industries to innovate safely, and build automated systems without the prohibitive computational costs associated with multi-pass semantic entropy approaches.

References

[1] J. Farquhar, J. Kenton, A. Mikulik, S. Marcus, and S. Legg, "Detecting hallucinations in language models using semantic entropy," *Nature*, 2024.

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[5] X. Wang et al., "Advancing LLM-Generated Code Reliability: A Hybrid Approach," *IEEE Transactions on Software Engineering*, 2025.

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[9] M. Niu, H. Haddadi, and G. Pang, "Robust Hallucination Detection via Adaptive Token Selection," *ICLR*, 2025.

[10] M. Yamada and Y. Arase, "Light-Weight Detection Using Contrastive Learning for Conditional Text Generation," *ACL*, 2025.

TABLE I
COMPARATIVE ANALYSIS OF HALLUCINATION DETECTION LITERATURE

Sl. No	Paper Title	Author(s)	Journal Name	Month & Year	Method / Technology Used	Advantages	Limitations
1	Detecting hallucinations in language models using semantic entropy [1]	J. Farquhar, J. Kenton, A. Mikulik, S. Marcus, S. Legg	Nature (Q1)	2024	Semantic Entropy (bidirectional entailment checks)	High confidence detection (AUROC 0.790) without task-specific training.	High computational overhead; fails on confidently wrong data.
2	Siren's Song in the AI Ocean: A Survey on Hallucination [2]	X. Li et al.	Computational Linguistics (Q1)	2025	Taxonomy review of fact-conflicting types.	Comprehensive categorization; shows advanced models achieve ∼87% on HaluEval.	Mitigation of hallucinations remains an open research challenge.
3	A Survey on Hallucination in LLMs: Principles, Taxonomy, Challenges [3]	Y. Huang et al.	ACM Computing Surveys (Q1)	2025	Comprehensive taxonomy across training/inference.	Aggregates detection metrics to guide future RAG designs.	Identifies inherent unreliability acting as a barrier to industrial innovation.
4	Detecting Hallucination Through Deep Internal Representation Analysis [4]	Z. Luan et al.	IJCAI (Q1)	2025	Internal representation extraction using linear probing.	Highly precise detection via Multi-Layer Perceptron (MLP).	Requires full access to the model's internal hidden states.
5	Advancing LLM-Generated Code Reliability: A Hybrid Approach [5]	X. Wang et al.	IEEE Trans. on Software Eng. (Q1)	2025	Static and dynamic hybrid code execution analysis.	Effectively identifies structural and logical coding hallucinations.	High architectural complexity for real-time applications.
6	A Review of Faithfulness Metrics for Hallucination Assessment [6]	A. Boulgouris et al.	IEEE J. Sel. Top. Signal Process. (Q1)	2025	Systematic review of semantic faithfulness metrics.	Clearly categorizes evaluation tools and benchmarks.	Lacks a unified metric applicable across all domains.
7	Hallucination Detection Using Spectral Features of Attention Maps [7]	F. Barbero et al.	EMNLP (Q1)	2025	Spectral feature analysis of attention matrix graphs.	Offers universal, white-box detection without domain adaptation.	Strictly limited to transformer-based architectures.
8	A Survey of Automatic Hallucination Evaluation [8]	S. Qi, L. Gui, Y. He, Z. Yuan	IEEE TNNLS (Q1)	2024	Automated evaluation of natural language generation.	Deep coverage of inconsistencies in generated outputs.	Places less emphasis on concrete mitigation frameworks.
9	Robust Hallucination Detection via Adaptive Token Selection [9]	M. Niu, H. Haddadi, G. Pang	ICLR (Q1)	2025	Adaptive token-level processing and selection.	Significantly reduces computational cost during inference.	Struggles to maintain accuracy on extremely long text contexts.
10	Light-Weight Detection Using Contrastive Learning [10]	M. Yamada, Y. Arase	ACL (Q1)	2025	Contrastive learning over conditional text generations.	High speed and efficiency suitable for conditional generation.	Heavily dependent on specialized negative training pairs.